CS6030 — Logic and Combinatorics for CS

Tier	C
Phase (order)	8
Do this first	CS1200, CS1202
Estimated effort	10-12 weeks
Unlocks	CS5692, CS6850

Roadmap

1. Propositional and first-order logic, resolution, SAT.
2. Advanced counting and probabilistic method.

Video courses (all links checked on YouTube on 2026-10-22)

🎓 Main course: follow it start to end

- Mathematics - Mathematical Logic (42 videos). NPTEL *Mathematical Logic* (nptelhrd): propositional/first-order logic, soundness, completeness.

🎥 Other full lecture series (pick one if the main one doesn't click, or for topics it skips)

- Computer Science - Combinatorics (41 videos). NPTEL *Combinatorics* (nptelhrd): counting, recurrences, generating functions, Polya.
- Set Theory and Mathematical Logic (52 videos). NPTEL *Set Theory and Mathematical Logic* (IIT Kanpur).
- MIT 18.200 Principles of Discrete Applied Mathematics, Spring 2024 (20 videos). MIT 18.200 *Principles of Discrete Applied Mathematics* (Spring 2024, official OCW).

Study plan (week by week)

Lecture numbers are positions in the main playlist above. Click a number to open that video.

Week	Lectures	What to do
1	1–6	Propositional logic: syntax, unique parsing, semantics, consequence.
2	7–14	Informal proofs, normal forms, SAT/3SAT, Horn-SAT, resolution. Write a DPLL SAT solver; try PySAT/MiniSat.
3	15–21	Propositional calculus, adequacy, compactness, analytic tableaux.
4	22–31	First-order logic: syntax, semantics, validity, quantifier laws, prenex form.
5	32–38	Skolem form, Herbrand's theorem, unification, resolution. Implement unification.

Week	Lectures	What to do
6	39–42	Axiomatic system, semi-decidability, tableaux for FOL, Gödel's incompleteness. Combinatorics half: use the alt series.

▼ **All lectures in *Mathematics - Mathematical Logic* (42 videos)**

1. Mod-01 Lec-01 Sets and Strings
2. Mod-01 Lec-02 Lecture-02-Syntax of Propositional Logic
3. Mod-01 Lec-03 Lecture-03-Unique Parsing
4. Mod-01 Lec-04 Lecture-04-Semantics of PL
5. Mod-01 Lec-05 Lecture-05-Consequences and Equivalences
6. Mod-01 Lec-06 Five results about PL
7. Mod-01 Lec-07 Lecture-07-Calculations and Informal Proofs
8. Mod-01 Lec-08 Lecture-08-More Informal Proofs
9. Mod-01 Lec-09 Lecture-09-Normal forms
10. Mod-01 Lec-10 Lecture-10-SAT and 3SAT
11. Mod-01 Lec-11 Lecyture-11-Horn-SAT and Resolution
12. Mod-01 Lec-12 Lecture-12-Resolution
13. Mod-01 Lec-13 Lecture-13-Adequacy of Resolution
14. Mod-01 Lec-14 Lecture-14-Adequacy and Resolution Strategies
15. Mod-01 Lec-15 Lecture-15-Propositional Calculus (PC)
16. Mod-01 Lec-16 Lecture-16-Some Results about PC
17. Mod-01 Lec-17 Lecture-17-Arguing with Proofs
18. Mod-01 Lec-18 Lecture-18-Adequacy of PC
19. Mod-01 Lec-19 Lecture-19-Compactness & Analytic Tableau
20. Mod-01 Lec-20 Lecture-20-Examples of Tableau Proofs
21. Mod-01 Lec-21 Lecture-21-Adequacy of Tableaux
22. Mod-01 Lec-22 Lecture-22-Syntax of First order Logic (FL)
23. Mod-01 Lec-23 Lecture-23-Symbolization & Scope of Quantifiers
24. Mod-01 Lec-24 Lecture-24-Hurdles in giving Meaning
25. Mod-01 Lec-25 Lecture-25-Semantics of FL
26. Mod-01 Lec-26 Lecture-26-Relevance Lemma
27. Mod-01 Lec-27 Lecture-27-Validity, Satisfiability & Equivalence
28. Mod-01 Lec-28 Lecture-28-Six Results about FL
29. Mod-01 Lec-29 Lecture-29-Laws, Calculation & Informal Proof
30. Mod-01 Lec-30 Lecture-30-Quantifier Laws and Consequences
31. Mod-01 Lec-31 Lecture-31-More Proofs and Prenex Form
32. Mod-01 Lec-32 Lecture-32-Prenex Form Conversion
33. Mod-01 Lec-33 Lecture-33-Skolem Form
34. Mod-01 Lec-34 Lecture-34-Syntatic Interpretation
35. Mod-01 Lec-35 Lecture-35-Herbrand's Theorem

36. Mod-01 Lec-36 Lecture-36-Most General Unifiers
37. Mod-01 Lec-37 Lecture-37-Resolution Rules
38. Mod-01 Lec-38 Lecture-38-Resolution Examples
39. Mod-01 Lec-39 Lecture-39-Ariomatic System FC
40. Mod-01 Lec-40 Lecture-40-FC and Semidecidability of FL
41. Mod-01 Lec-41 Lecture-41-Analytic Tableau for FL
42. Mod-01-Lec-42 Lecture-42-Godels Incompleteness Theorems

Books, notes and websites

- Book: Huth & Ryan, *Logic in Computer Science*
- Book: van Lint & Wilson, *A Course in Combinatorics*

Practice / proof of understanding

Write a DPLL SAT solver.

What your IITM course sheet says

From the PDF: CS6030

Official description: The course is intended to provide the foundations to mathematical rigor that are used in most areas of Computer Science and limitations of computation. The topics include Logic, Basic Combinatorics and elements of Computability.

Official syllabus (trimmed):

> Language of Math – Sets, relations, functions (quick review) Logic : Propositional Logic, Predicate and First order Logic, Examples, Soundness and Completeness. Second Order Logic. Incompleteness theorems. Further extensions of logic (MSO, Temporal etc). Subsets of first order logic. Example Applications. Proofs: Mathematical Arguments and Argument Forms, Inference Rules, Notion of a Proof and validity of an argument, Various Proof techniques and examples, Identifying fallacies. Infinite Sets: Countable and uncountable sets, Cantor's diagonalization. Undecidability : Turing Machines, Church-Turing Thesis. Notion of Computation and decidability, Undecidability of the Halting Problem. Consequences to the Program Verification Problem. Combinatorics Basics: Pigeonhole Principle and applications (e.g., Ramsey Theorem) Counting methods: Principle of Inclusion Exclusion, Proving Combinatorial Identities, Combinatorial Arguments, Permutations, Derangements. Recurrence: Linear recurrences, method of roots. Generating Functions. Examples Structured Sets: Posets and Lattices, Fixed Point Theorems (Knaster-Tarski and Polya), Monoids, Semigroups, Groupoids and Groups, Examples, Subgroups, Cosets, Lagrange's theorem. Introduction to Polya's theory of counting.

Official textbooks: 1) Discrete Mathematics and its Applications – Kenneth H. Rosen 7th Edition -Tata McGraw Hill Publishers – 2007.2) Introduction to Automata Theory, Languages and Computation (third edition) - John Hopcroft, Rajeev Motwani, and Jeffrey Ullman, Pearson Education Publishers, 2006.3) A Mathematical Introduction to Logic (second edition) - Herbert B Enderton, Academic Press , 2001.

Official references: 1. Combinatorics: Topics, Techniques, Algorithms by Peter J. Cameron, Cambridge University Press, 1994 (reprint 1996).2. Concrete Mathematics – Ronald Graham, Donald Knuth, and Oren

Patashnik, 2nd Edition - Pearson Education Publishers - 1996.

From the PDF: CS5692

Official description: Mathematical Logic forms the foundation of many theoretical and practical aspects of computer science. This course aims to introduce the students to the foundations of Mathematical Logic and its applications in computer science.

Official syllabus (trimmed):

> Module 1:Propositional Logic (PL) and First Order Logic (FOL) [3 Weeks] Syntax Structures and Semantics Modeling various scenarios in these logics, Logical Paradoxes Notions of Satisfiability, Model Checking, Validity Compactness Theorem Module 2 : Axiom Systems for PL and FOL [2 Weeks] Hilbert style axiom system Natural Deduction Sequent Calculus Module 3 : Modal Logic (ML) [3 Weeks] Syntax Kripke Structures and Semantics Epistemic Logic, Temporal Logic, Doxastic Logic Sound and Complete Axiom System Bisimulation Module 4 : Algorithms and Complexity [3 Weeks] Satisfiability problem for PL, FOL and ML Model Checking problem for PL, FOL and ML Decidable fragments of FOL (Un)decidability of various Theories Module 5 : Applications (Some of the following topics depending on time) [3 Weeks] Logic and Databases Logic and Complexity Logic and Automata Theory Logic and Formal Verification Logic and Artificial Intelligence

Official textbooks: A Mathematical Introduction to Logic by Herbert Enderton Publication : Academic Press, Year : 2001 Modal Logic by Patrick Blackburn, Maarten de Rijke and, Yde Venema Publisher : Cambridge University Press, 2nd Edition ; Year : 2014

Official references: Dynamic Epistemic Logic by Hans Ditmarsch , Wiebe Hoek , Barteld Kooi Publication : Academic Press, Year : 2001 Principles of Databases by Marcelo Arenas, Pablo Barcelo ´, Leonid Libkin, Wim Marten and Andreas Pieris Publication : Springer Year : 2008 The Classical Decision Problem by Egon Borger, Erich Gradel and Yuri Gurevic Publication : Springer Year : 1997 Logic and Automata: History an